Thermo-hydraulic optimization of a NEPCM micro-channel with a teardrop-shaped obstacle

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Abstract

This study assesses a passive teardrop-shaped obstacle and nano-encapsulated phase change materials (NEPCM) in a heated micro-channel. A validated Fluent model, with NEPCM properties implemented in a UDF, is coupled to an optiSLang optimization of 100 designs. The optimum, validated with Fluent, gives $PEC = 0.865$: the obstacle adds about 6 % heat transfer but multiplies the pressure drop by 1.7 to 4.

1 Problem setup

The domain is a 2D planar channel of length $L = 10$ mm and height $H = 1$ mm ($D_h = 2H = 2$ mm). An adiabatic teardrop-shaped obstacle (radius $R = 0.125$ mm, tip 0.25 mm downstream of the centre) is placed at $(3; 0)$ mm. Its shape is controlled by two stretch factors: s_x along the flow and s_y across it. The domain is shown in Fig. 1 and the boundary conditions in Table 1.

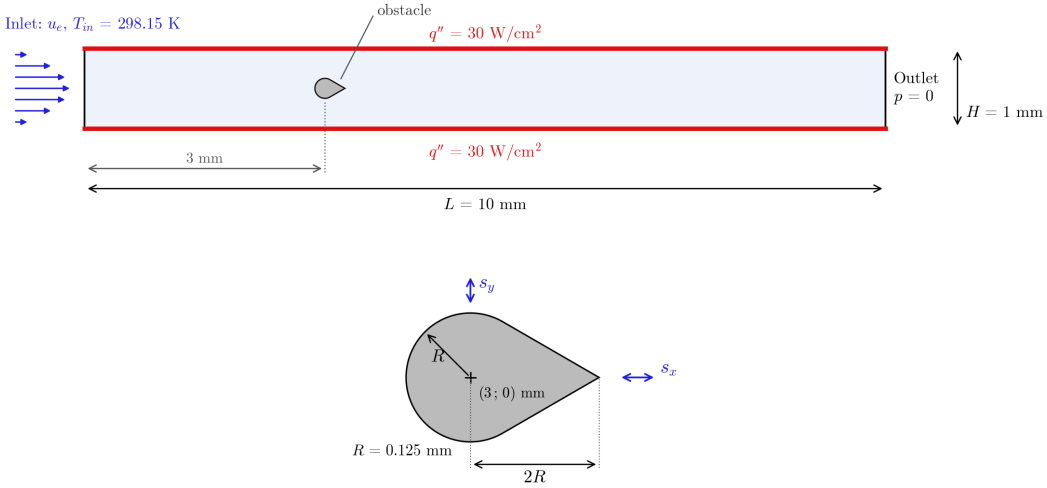


Figure 1: Computational domain and boundary conditions (top); obstacle geometry (bottom).

Table 1: Boundary conditions.

Boundary	Condition
Inlet	parabolic profile of mean velocity u_e, $T_{in} = 298.15$ K
Outlet	$p = 0$ Pa, backflow $T = 298.15$ K
Top and bottom walls	uniform heat flux $q'' = 30$ W/cm^2
Obstacle	adiabatic, no slip

Performance is measured at equal velocity against a smooth channel (subscript 0):

$$PEC = \frac{Nu/Nu_0}{(\Delta p/\Delta p_0)^{1/3}}, \qquad Nu = \frac{q''\, D_h}{k_w\,(\overline{T}_w - T_{in})}. \tag{1}$$

2 Governing equations and property models

The suspension is treated as a single homogeneous fluid. The flow is steady, laminar and incompressible:

$$\nabla \cdot \mathbf{u} = 0, \tag{2}$$

$$\rho\,(\mathbf{u}\cdot\nabla)\,\mathbf{u} = -\nabla p + \nabla\cdot(\mu\,\nabla\mathbf{u})\,, \tag{3}$$

$$\rho\,C_p(T)\,(\mathbf{u}\cdot\nabla T) = \nabla\cdot(k\,\nabla T)\,. \tag{4}$$

The mixture properties (subscripts w: water, p: capsule; φ: volume fraction) are

$$\rho = (1-\varphi)\,\rho_w + \varphi\,\rho_p, \tag{5}$$

$$k = k_w\,\frac{k_p + 2k_w - 2\varphi\,(k_w - k_p)}{k_p + 2k_w + \varphi\,(k_w - k_p)} \qquad \text{(Maxwell [1])}, \tag{6}$$

$$\mu = \mu_w\left(1 + 2.5\,\varphi + 6.5\,\varphi^2\right) \qquad \text{(Batchelor [2])}. \tag{7}$$

Melting is represented by a heat capacity with a sinusoidal peak over the melting range T_{mr} around T_f [3]:

$$C_p(T) = \begin{cases} \omega\,C_{p,p} + (1-\omega)\,C_{p,w} + \omega\,\dfrac{\pi}{2}\left(\dfrac{h_{sf}}{T_{mr}} - C_{p,p}\right)\sin\!\left(\pi\,\dfrac{T - T_f + T_{mr}/2}{T_{mr}}\right) & \text{if } |T-T_f| \le \dfrac{T_{mr}}{2}, \\ \omega\,C_{p,p} + (1-\omega)\,C_{p,w} & \text{otherwise,} \end{cases} \tag{8}$$

where $\omega = \varphi\,\rho_p/\rho$ is the capsule mass fraction. Material data are given in Table 2.

Table 2: Material properties (capsule with n-octadecane core [4]).

Property	Water	Capsule
ρ (kg/m^3)	997.1	1033
k (W/m.K)	0.613	0.2
C_p (J/kg.K)	4179	1830
μ (Pa.s)	7.97×10^{-4}	–
h_{sf} (J/kg)	–	152 600
T_f ; T_{mr} (K)	–	301.15 ; 0.5

Implementation. Equations (5) to (8) are implemented in a user-defined function (UDF). Equations (2) to (4) are solved with Ansys Fluent 2026 R1 (pressure-based coupled solver, second-order schemes, residuals below 10^{-6}).

3 Mesh

A structured O-grid surrounds the obstacle and is refined towards the walls and the obstacle surface (Fig. 2). The obstacle shape is changed by mesh morphing, so every design keeps the same topology. Mesh quality is given in Table 3.

Table 3: Quality of the medium mesh.

Metric	Value
Number of cells	31 200
Minimum orthogonal quality	0.494
Maximum aspect ratio	13.4

4 Mesh convergence (GCI)

Three meshes with a constant refinement ratio $r = 1.5$ were compared on the case $\varphi = 0.04$, $s_x = 1$, $s_y = 1.4$, $u_e = 0.6$ m/s, following the ASME procedure [5] (Table 4).

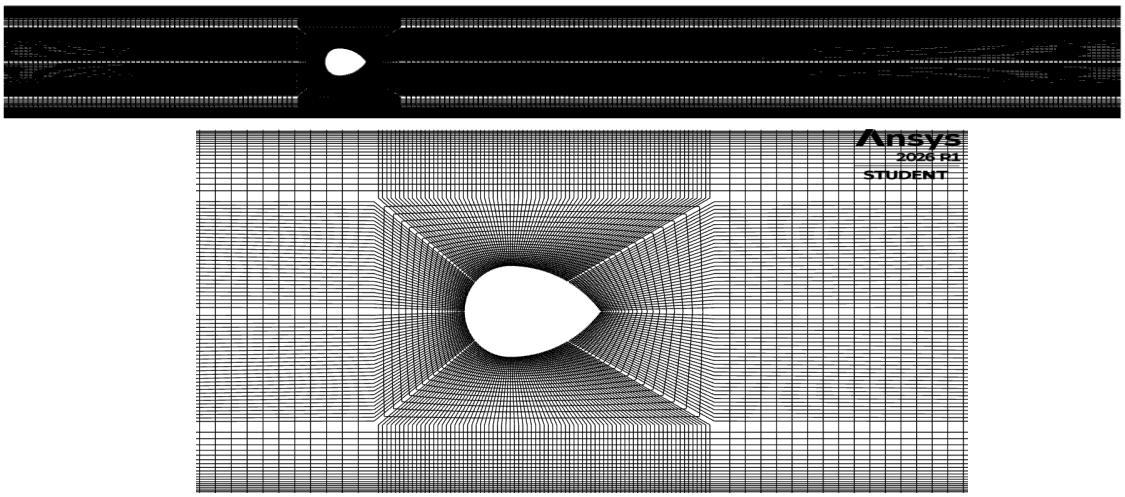


Figure 2: Medium mesh (31 200 cells): full channel (top) and O-grid around the obstacle (bottom).

Table 4: Spatial discretization error.

Parameters	$\psi = Nu_m$ (monotonic convergence)	$\psi = \Delta P$ (monotonic convergence)
N_1 ; N_2 ; N_3	70 410 ; 31 200 ; 13 780	
r_m	1.5	
ψ_1 ; ψ_2 ; ψ_3	23.871 ; 23.843 ; 23.781	177.71 ; 178.02 ; 179.06
O_a	1.96	2.99
ψ_{ext}^{21}	23.894	177.578
E_a^{21}	0.117 %	0.174 %
E_{ext}^{21}	0.097 %	0.074 %
GCI_{fine}^{21}	0.12 %	0.09 %

Convergence is monotonic with an order close to 2, as expected for second-order schemes. The medium mesh is retained (0.1–0.2 % from the fine mesh).

5 Verification and validation

Table 5: Comparison with analytical references (200 mm channel, fully developed flow).

Quantity	Present study	Reference	Deviation
Nu	8.310	8.235 [6]	0.92 %
Δp	190.85 Pa	191.28 Pa [6]	0.22 %

The reference values Nu_0 and Δp_0 in Eq. (1) are computed on the smooth channel (11 runs between 0.1 and 0.6 m/s).

6 Optimization setup

The design space is given in Table 6. The objective is to maximize PEC under the constraint $T_{max} \leq$ 363.15 K (90 °C). optiSLang AMOP [7] builds a metamodel from 100 advanced Latin hypercube

designs; all of them converged and 47 satisfy the constraint.

Table 6: Design parameters.

Parameter	Range	Role
φ	0 – 0.04	NEPCM volume fraction
s_x	0.8 – 1.8	obstacle length
s_y	0.6 – 1.4	obstacle height (15 to 35 % blockage)
u_e	0.1 – 0.6 m/s	inlet velocity (Re from 250 to 1 440)

Metamodel quality and parameter influence are measured by the optiSLang coefficient of prognosis (CoP, Table 7).

Table 7: optiSLang coefficients of prognosis (CoP). "–": parameter filtered out as negligible.

Output	φ	s_x	s_y	u_e	Total CoP
PEC	0.039	0.047	0.560	0.401	0.993
T_{max}	–	0.021	0.022	0.978	0.989
Nu	–	–	–	0.985	0.985
Δp	–	–	0.158	0.861	0.984

The metamodel is highly reliable (total CoP above 0.98). PEC is driven mainly by the obstacle height s_y, then by the inlet velocity u_e; T_{max} depends almost only on u_e. The NEPCM fraction and the obstacle length have a minor influence.

7 Results and discussion

7.1 Heat transfer gain and hydraulic cost

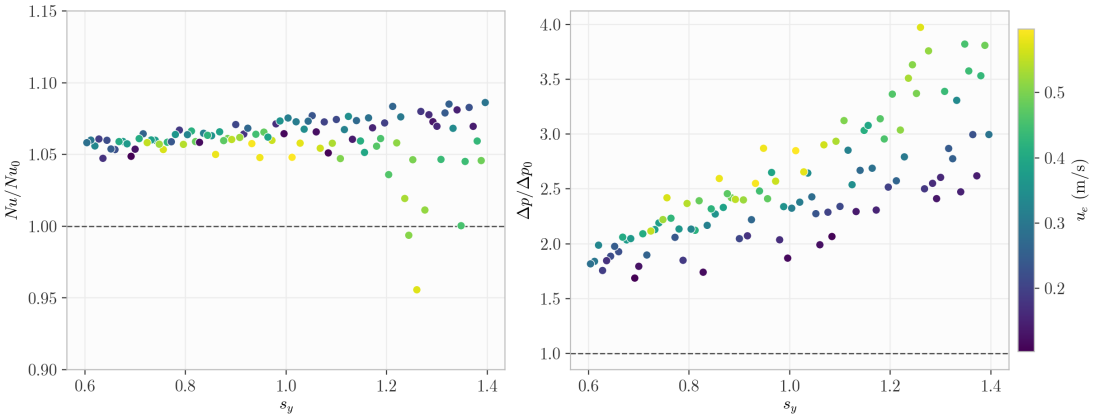


Figure 3: Heat transfer gain Nu/Nu_0 and hydraulic cost $\Delta p/\Delta p_0$ of the 100 designs versus obstacle height.

The obstacle raises Nu by only about 6 % (Nu/Nu_0 between 1.05 and 1.09), almost independently of its height, whereas the pressure drop is multiplied by 1.7 to 4 and grows clearly with s_y (Fig. 3). PEC is therefore governed by the hydraulic penalty. A few tall and short obstacles at high velocity are exceptions ($Nu/Nu_0 < 1$); this case is analysed in Section 7.4.

7.2 Thermal constraint and optimum

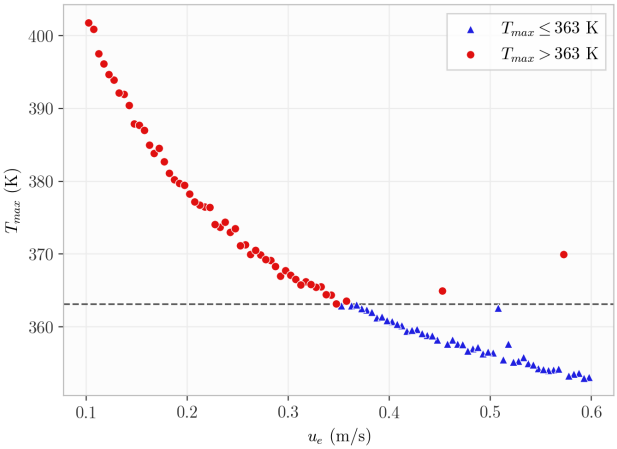


Figure 4: Maximum temperature of the 100 designs versus inlet velocity.

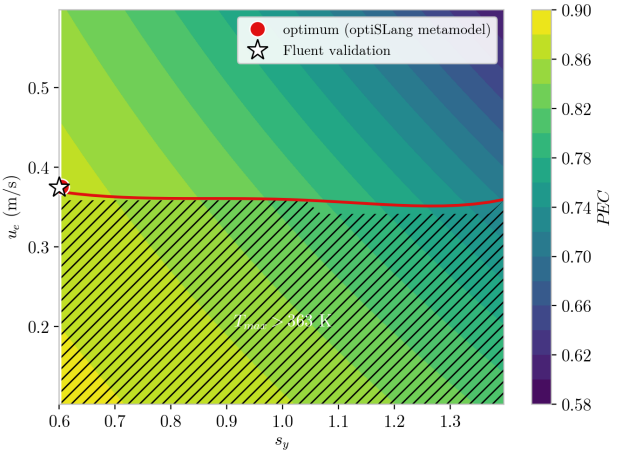


Figure 5: PEC response surface from the optiSLang metamodel ($\varphi = 0$, $s_x = 1.5$). Hatched: $T_{max} > 363$ K.

T_{max} decreases with u_e, and the 363 K limit requires a minimum inlet velocity of about 0.35 m/s (Fig. 4). PEC alone would be misleading: the best PEC of the sample (0.881, at $u_e = 0.10$ m/s) comes with a 402 K wall.

φ and s_x, which have little influence (Table 7), are set to 0 and 1.5. The optimum lies at the minimum obstacle height, right on the thermal limit (Fig. 5). It was recomputed with Fluent (Table 8).

Table 8: Optimum: optiSLang prediction and Fluent computation ($\varphi = 0$, $s_x = 1.5$, $s_y = 0.6$, $u_e = 0.375$ m/s).

Output	optiSLang	Fluent	Deviation
PEC	0.868	0.865	$-0.4\,\%$
T_{max} (K)	362.8	362.0	-0.8 K
Nu	–	20.81	–
Δp (Pa)	–	65.46	–
R_{th} (K.cm^2/W)	–	2.130	–

The Fluent computation confirms the prediction within 0.4 % and satisfies the thermal constraint.

7.3 Flow and thermal fields

Figures 6 and 7 compare three cases at $u_e = 0.345$ m/s and $\varphi = 0.04$: smooth channel, baseline obstacle ($s_x = 1$, $s_y = 1.4$) and optimal shape ($s_x = 1.5$, $s_y = 0.6$).

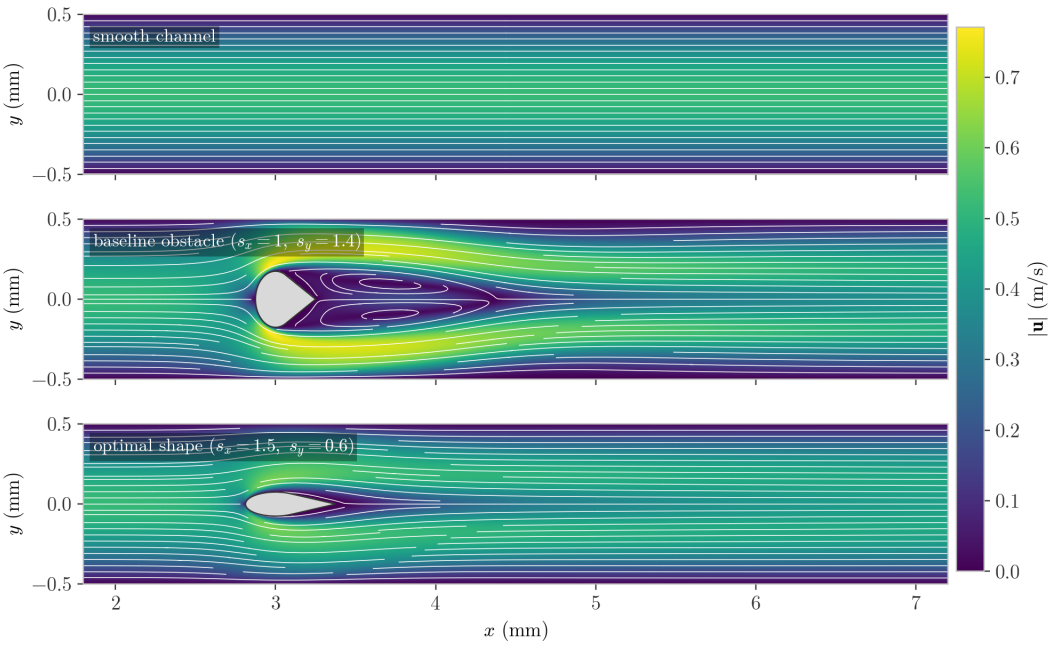


Figure 6: Velocity magnitude and streamlines.

The baseline obstacle accelerates the flow in the two gaps and creates a long recirculation zone in its wake (Fig. 6). The optimal shape disturbs the flow much less, hence its lower pressure drop.

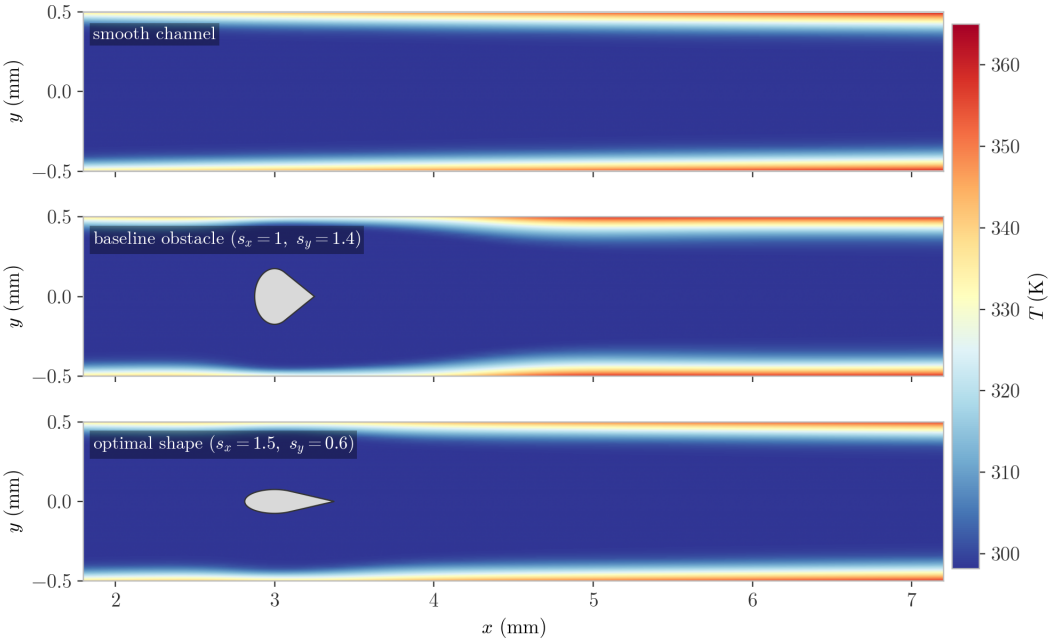


Figure 7: Temperature field.

Heat remains confined in thin layers along the walls, while the core stays at the inlet temperature (Fig. 7). The obstacle thins these layers only locally.

7.4 Wall separation and hot spot (design DP83)

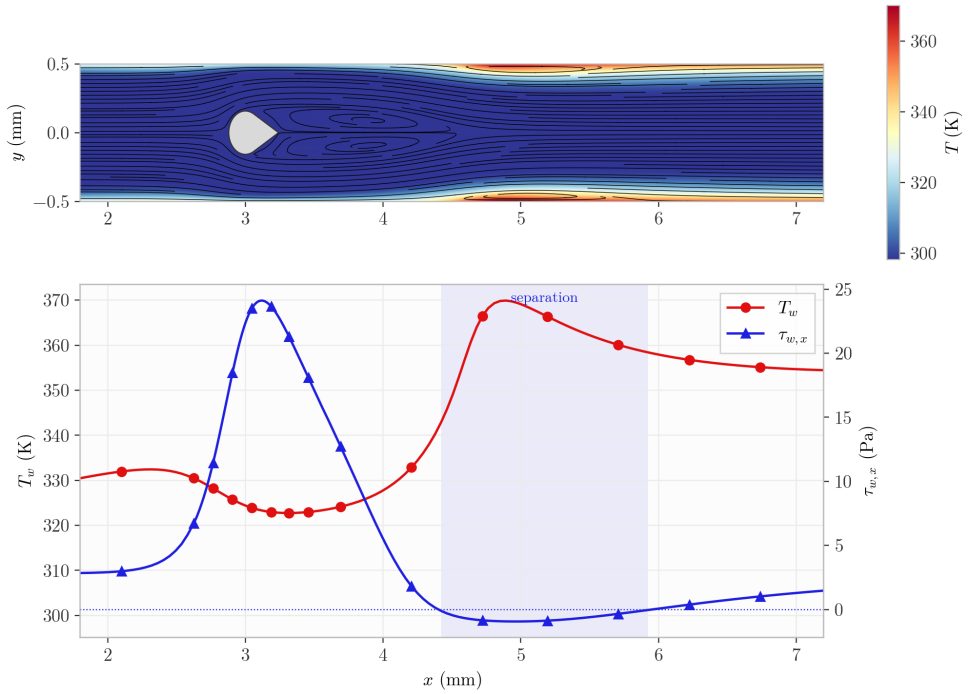


Figure 8: Design DP83 ($\varphi = 0.0114$, $s_x = 0.965$, $s_y = 1.26$, $u_e = 0.5725$ m/s): temperature and streamlines (top); bottom-wall temperature and wall shear stress (bottom).

Design DP83 reaches $T_{max} = 370$ K, about 15 K above the trend at this velocity. Fig. 8 shows the cause: downstream of the minimum flow section, the flow decelerates and separates from the wall. The wall shear stress becomes negative between $x = 4.4$ and 5.9 mm, and the temperature peak (370 K at $x = 4.9$ mm) lies inside this separation bubble. Tall and short obstacles therefore create a hot spot on the wall.

8 Conclusions

1. A passive centred obstacle does not pay off at equal velocity: the validated optimum gives $PEC = 0.865$.

2. The obstacle height controls the trade-off: it sets the pressure drop without improving heat transfer.

3. PEC must be paired with a temperature limit; without it, the best PEC comes with a 129 °C wall.

4. The NEPCM fraction has a small influence on PEC and none on T_{max}.

5. Tall and short obstacles cause wall separation and a local hot spot.

References

[1] J. C. Maxwell. *A Treatise on Electricity and Magnetism*. Clarendon Press, Oxford, 1873.

[2] G. K. Batchelor. The effect of Brownian motion on the bulk stress in a suspension of spherical particles. *Journal of Fluid Mechanics*, 83(1):97–117, 1977. doi:10.1017/S0022112077001062.

[3] M. Ghalambaz, A. J. Chamkha, D. Wen. Natural convective flow and heat transfer of nano-encapsulated phase change materials (NEPCMs) in a cavity. *International Journal of Heat and Mass Transfer*, 138:738–749, 2019.

[4] M. Chen, Y. Wang, Z. Liu. Experimental study on micro-encapsulated phase change material slurry flowing in straight and wavy microchannels. *Applied Thermal Engineering*, 190:116841, 2021. doi:10.1016/j.applthermaleng.2021.116841.

[5] I. B. Celik, U. Ghia, P. J. Roache, C. J. Freitas, H. Coleman, P. E. Raad. Procedure for estimation and reporting of uncertainty due to discretization in CFD applications. *Journal of Fluids Engineering*, 130(7):078001, 2008. doi:10.1115/1.2960953.

[6] R. K. Shah, A. L. London. *Laminar Flow Forced Convection in Ducts*. Academic Press, New York, 1978.

[7] T. Most, J. Will. Metamodel of Optimal Prognosis – an automatic approach for variable reduction and optimal metamodel selection. *Weimarer Optimierungs- und Stochastiktage 5.0*, Weimar, 2008.